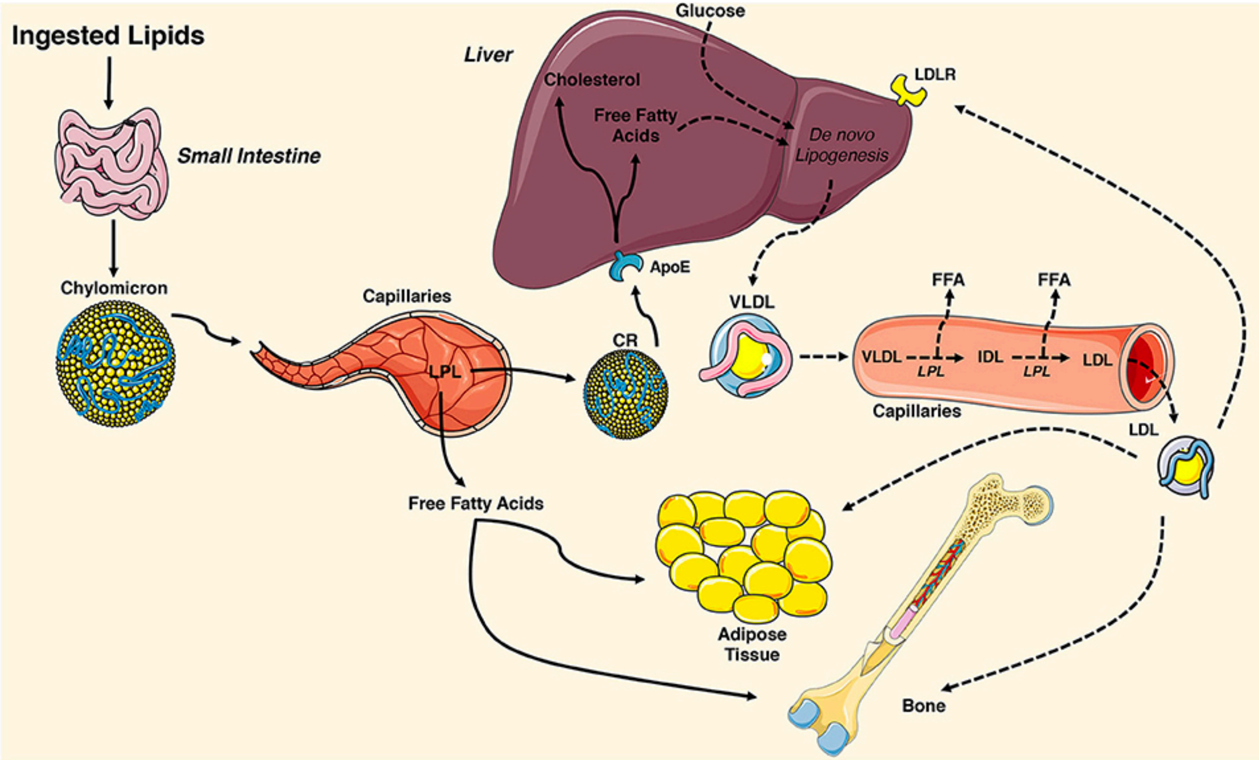
Lipid Transport



Alekos, Nathalie S., Megan C. Moorer, and Ryan C. Riddle. "Dual effects of lipid metabolism on osteoblast function." *Frontiers in Endocrinology* 11 (2020).

Dr. P. Satpati,
BSBE, IIT Guwahati

Before we start…
Few numerical question related the lipids.

A lipid-soluble organic compound is isolated from a sample. Elemental analysis gave 67.8% C, 9.60% H, and 22.6 % O. (a) What is the simplest empirical formula? (b) the minimum MW of the compound ?

(a) Divide the percent composition value by the atomic weight of the element.
gram-atoms of each element per 100 g of compound:
C= (67.8 /12) = 5.65 ; H= (9.6/1) = 9.6 ; O=22.6/16= 1.41

C:H:O = 5.65 : 9.6 : 1.41 ~ 4: 6.8 : 1

There must be whole number of atoms of each element.
C: H: O = 4x5 : 6.8x5 : 1x5 = 20 : 34 : 5

Thus formula: $C_{20}H_{34}O_5$

(b) Molecular weight (**$C_{20}H_{34}O_5$**) = 20x12+34x1 + 5x16 = **354**

The density of a homogeneous membrane (consist of protein & lipid) is 1.15 g/cm^3. The average density of proteins = 1.3 g/cm^3 and lipids = 0.92 g/cm^3. What are the weight fraction of protein and lipid in the membrane ?

Say Weight fraction of protein = x
Then, Weight fraction of lipid = (1-x)

Specific vol (V_{sp}^{Total}) = ∑(wt fraction of components) x (V_{sp} of component)

$V_{sp}^{Membrane}$ = (wt. fraction protein x $V_{sp}^{protein}$) + (wt. fraction lipid x V_{sp}^{lipid})

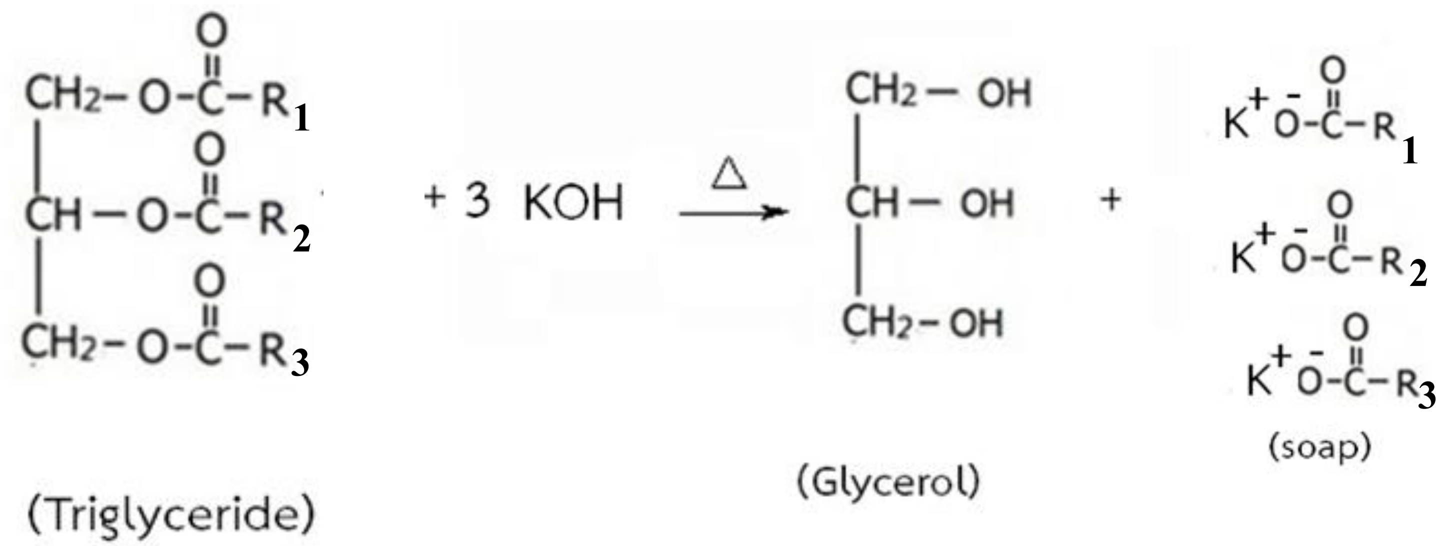
$V_{sp}^{Membrane} = (1/\ 1.15) = 0.87$; $V_{sp}^{protein} = (1/1.3) = 0.77$; $V_{sp}^{lipid} = (1/0.92) = 1.09$

$$0.87 = 0.77*x+(1-x)*1.09$$

=) x = 0.688 (PROTEIN)

Thus, 1-x = 0.312 (LIPID)

Calculate the saponification number of the following TRIGLYCERIDE.

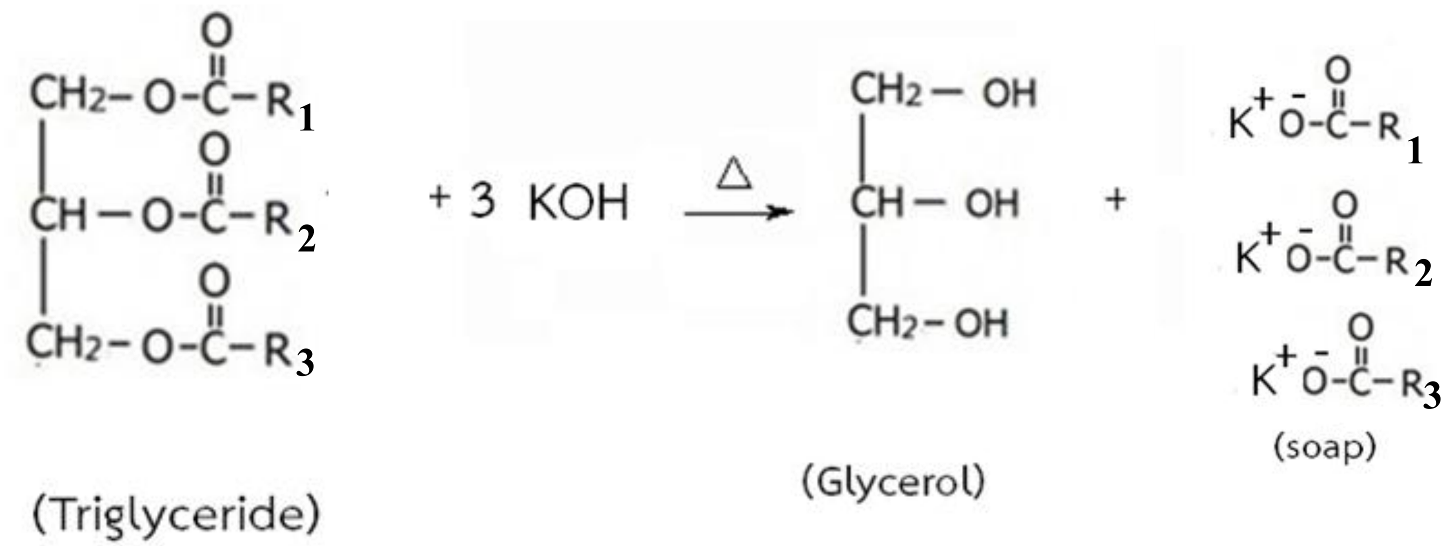


(Molecular Weight =862)

Saponification= *Process of making soap*

saponification number = *number of milligrams of potassium hydroxide required for complete hydrolysis of 1g of fat .*

TRIGLYCERIDE (Mol wt = 862)



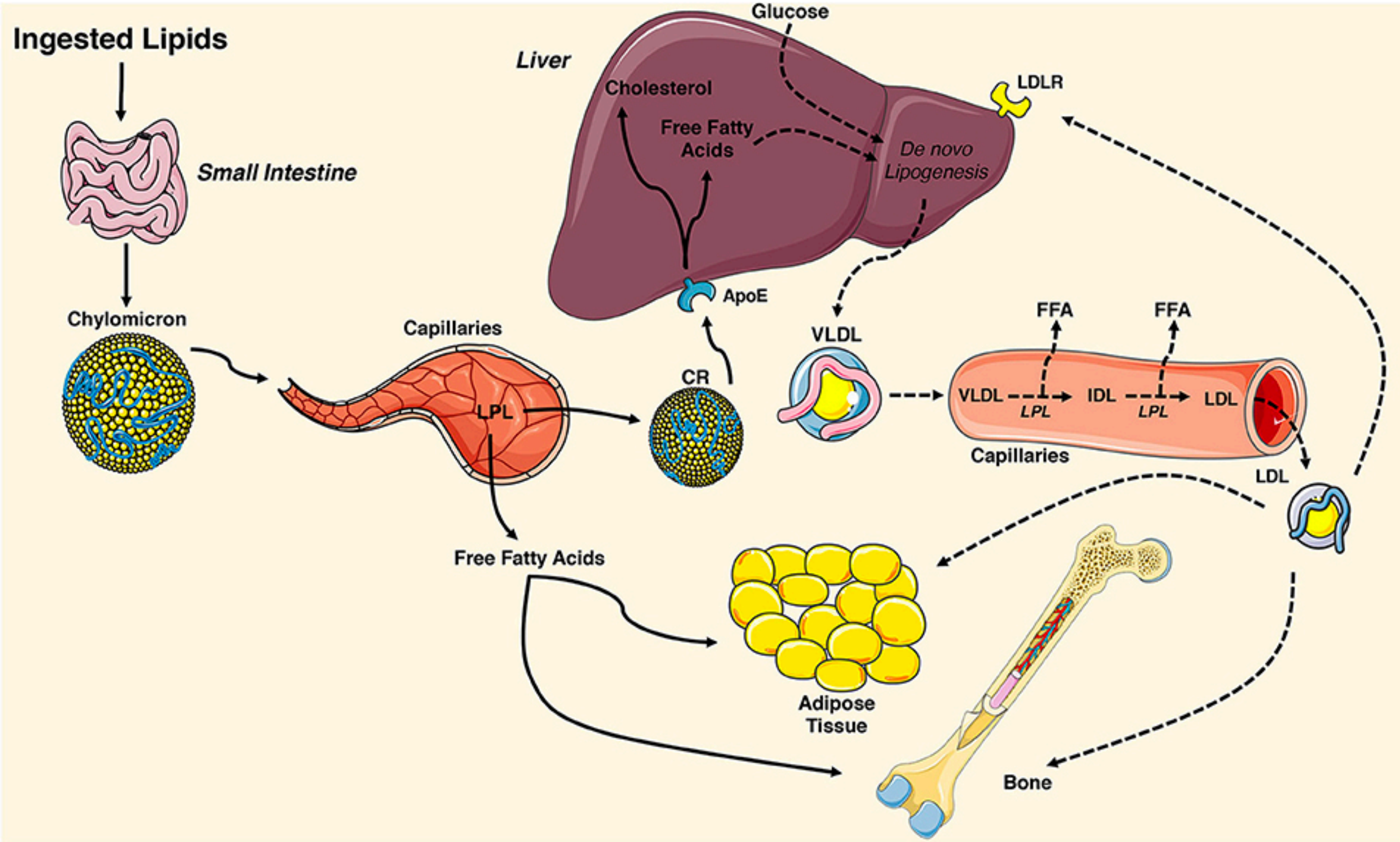
Three moles KOH is required to saponify one mol of TRIGLYCERIDE.

Mol weight of KOH = 56

Thus, 3*56 = 168 g of KOH required to saponify 862 g of triglyceride

Soponification number = (168 * 10^3 mg KOH) / (862 g triglyceride) = 194.9

Lipid Transport



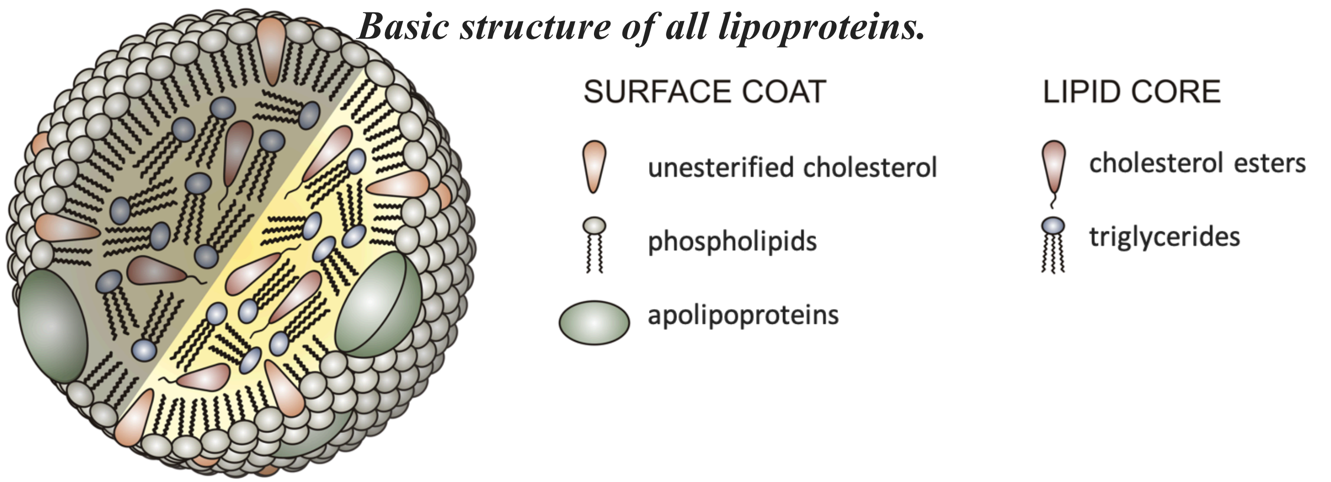
Alekos, Nathalie S., Megan C. Moorer, and Ryan C. Riddle. "Dual effects of lipid metabolism on osteoblast function." *Frontiers in Endocrinology* 11 (2020).

Lipids are Important

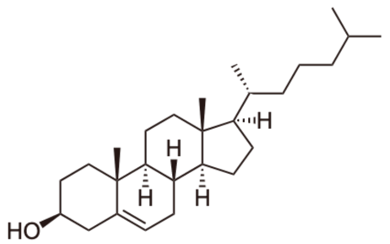
Source:
1. Food
2. Our body synthesize

Lipids are mostly non-polar (oily).
Thus, transport is water is tricky.
Transported as special packages called
LIPOPROTEINS

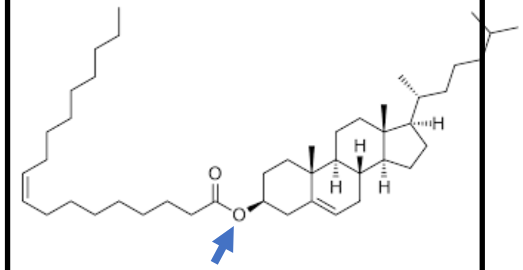
Basic structure of all lipoproteins.



https://openoregon.pressbooks.pub/nutritionscience/chapter/5e-lipid-transport-storage-util/

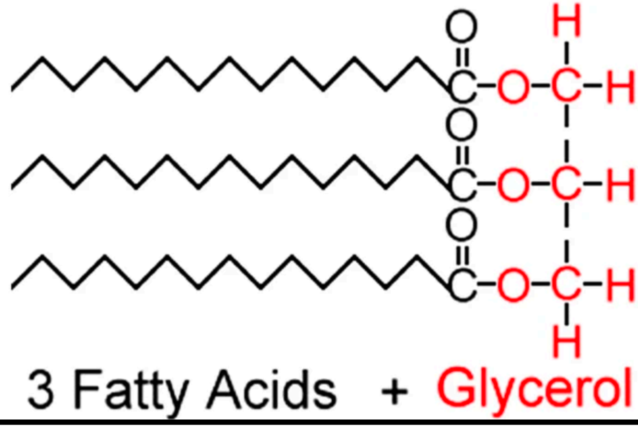


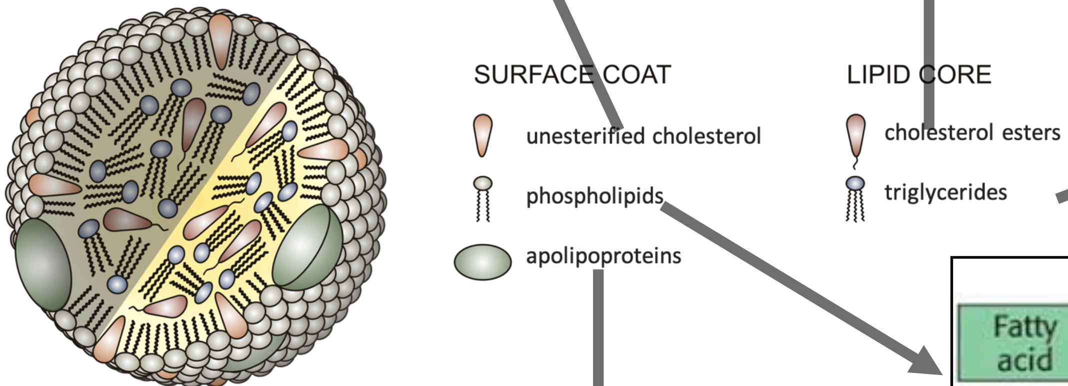
Cholesterol
(Lecture5, 2nd part, Slide18)



Cholesterol ester

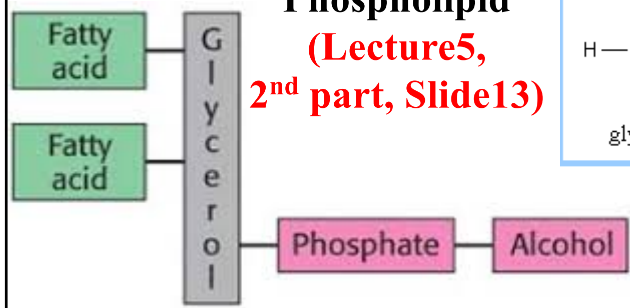
Triglyceride
(Lecture5, 2nd part, Slide 8)

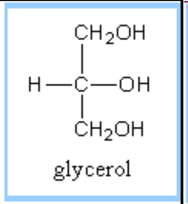


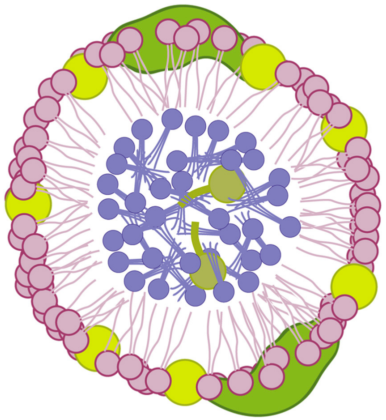


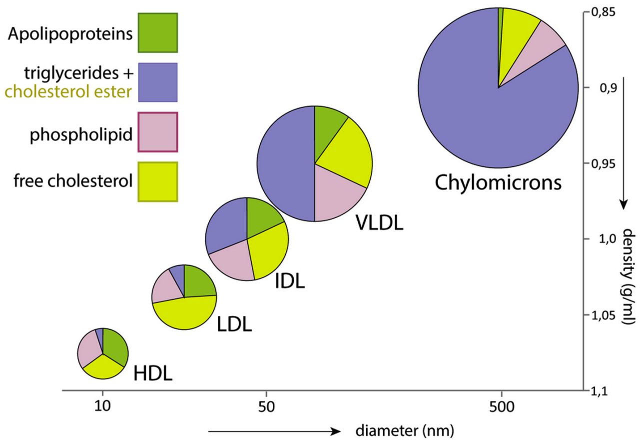
Proteins are one or multiple chains of amino acid residues. (B-48, B-100) **(Lecture5, 1st part, Slide 3)**

Phospholipid
(Lecture5, 2nd part, Slide13)

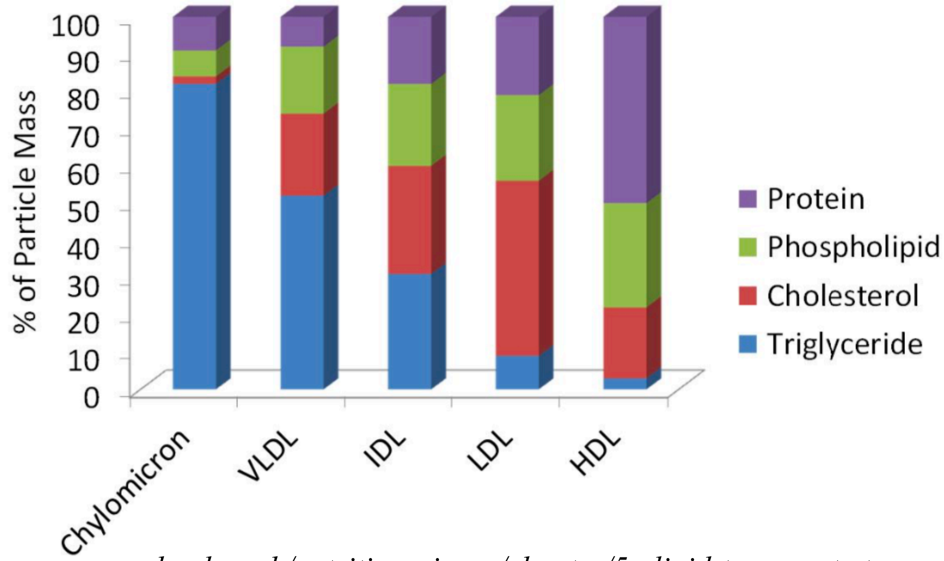








Progress in retinal and eye research 67 (2018): 56-86.



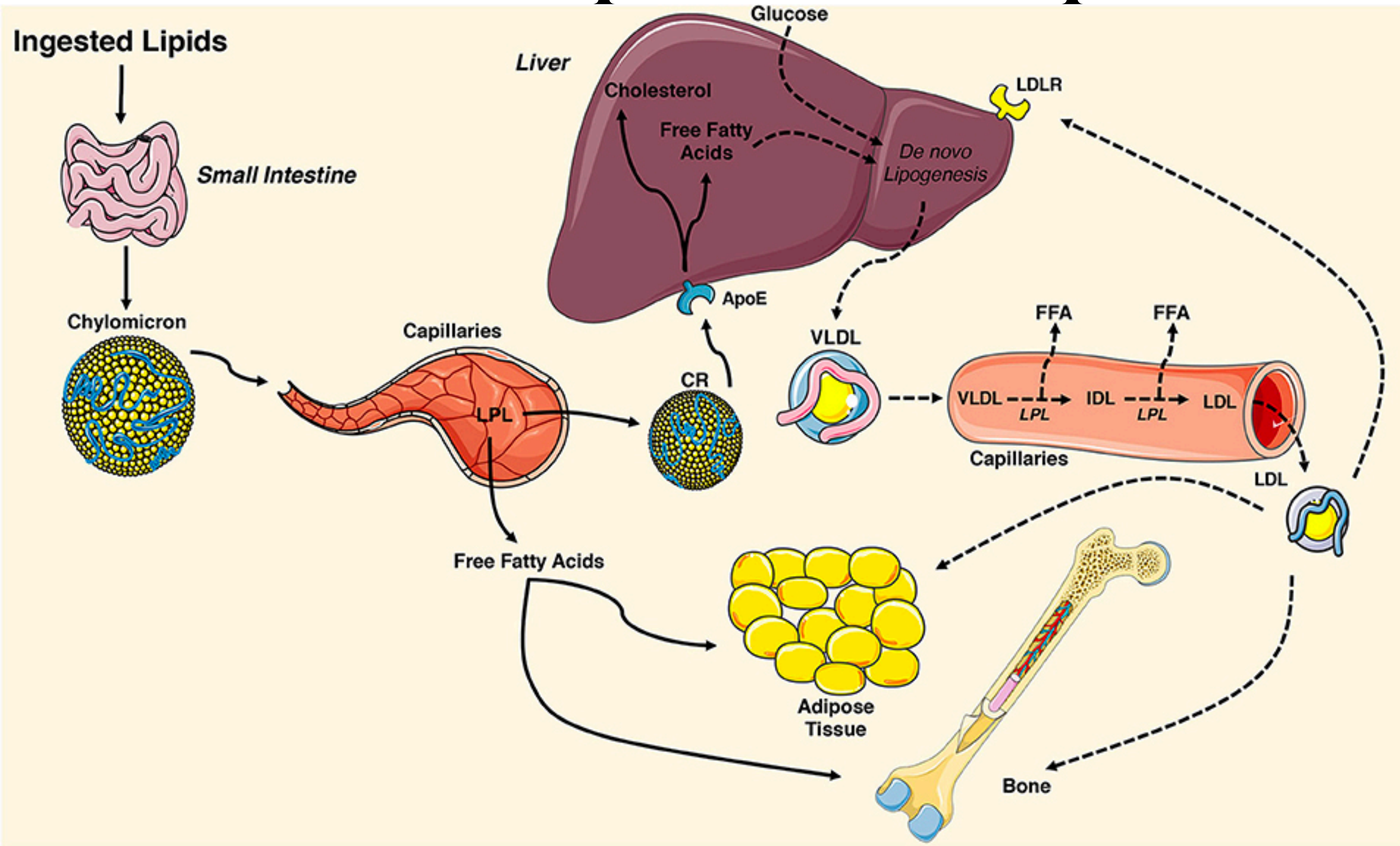
https://openoregon.pressbooks.pub/nutritionscience/chapter/5e-lipid-transport-storage-util/

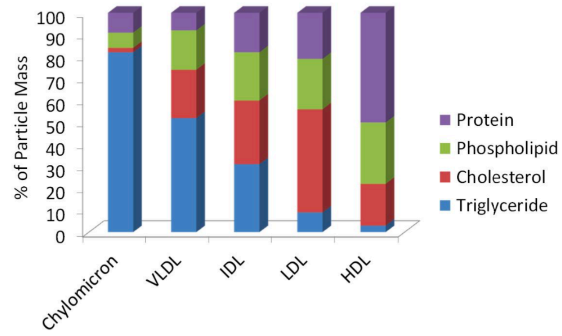
Based on the density, there are
five classes of lipoproteins (from high to low)

1. High-density lipoprotein (HDL),
2. Low-density lipoprotein (LDL),
3. Intermediate density lipoprotein (IDL),
4. Very-low-density lipoprotein (VLDL),
5. Chylomicron (CM).

- Protein content increases as size decreases
- LDL has highest concentration of Cholesterol

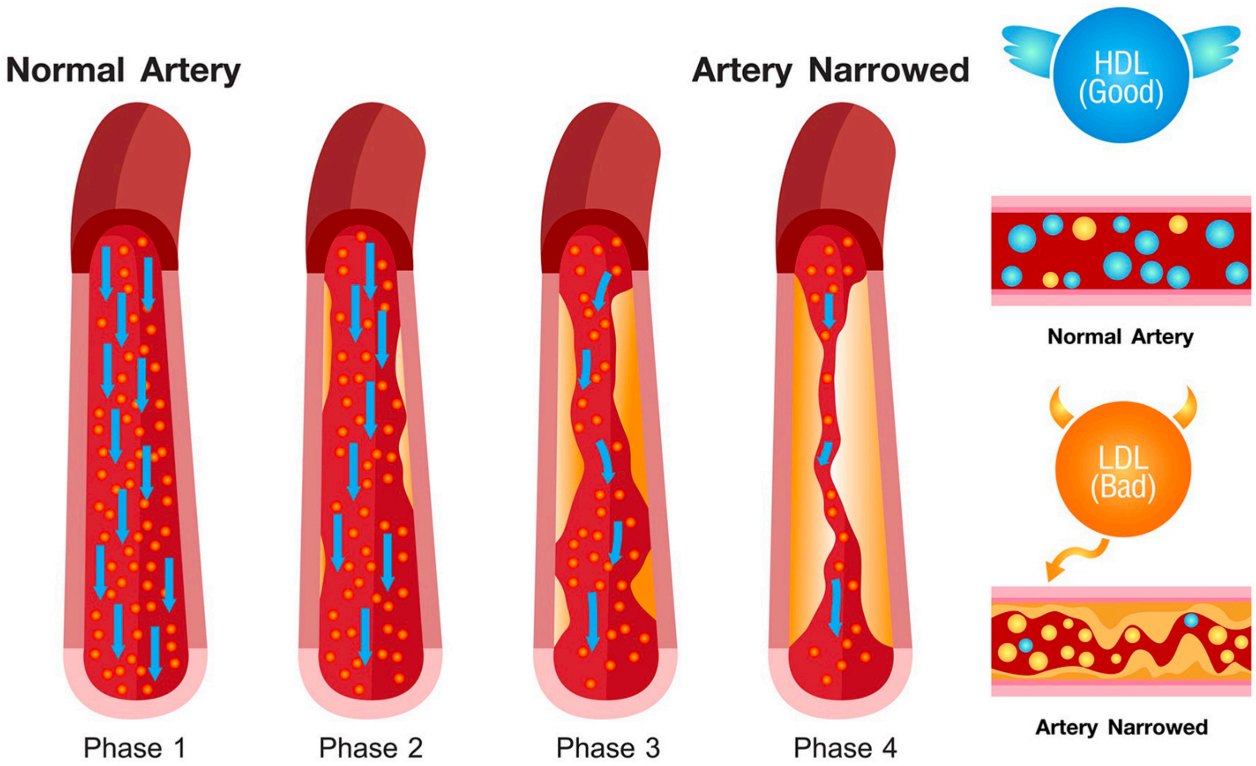
Lipid Transport





Alekos, Nathalie S., Megan C. Moorer, and Ryan C. Riddle. "Dual effects of lipid metabolism on osteoblast function." *Frontiers in Endocrinology* 11 (2020).

Cholesterol



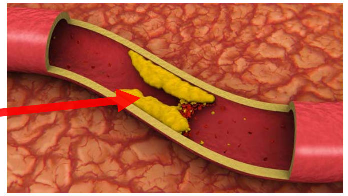
https://www.youtube.com/watch?v=9dghtf7Z7fw

Take home

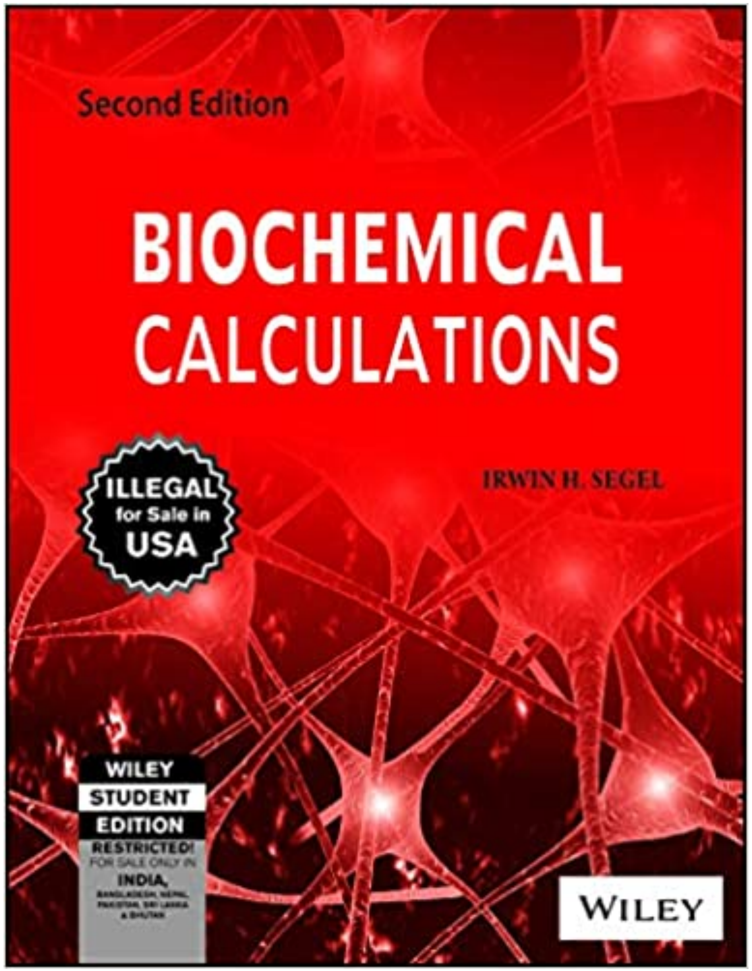
- **Lipids** are transported as lipo-protein (complex of protein and lipids).

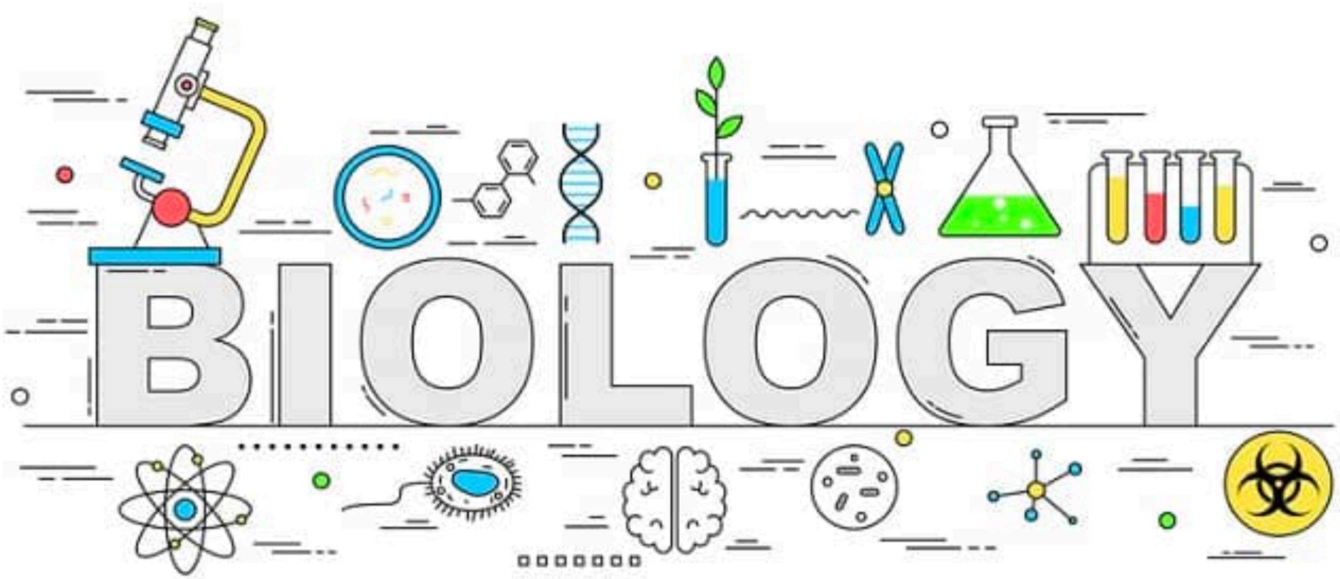
- 5 different types of lipid transport:
  1. Chylomicron (CM)
  [Food →Small intestine …….→ Liver]
  2. VLDL, LDL
  [De novo lypo-genesis(DNL)+food → body →Liver]
  3. HDL (produced by both intestine and liver)
  [Excess cholesterol in the blood is collected and delivered it back to liver]

- **LDL is bad** (Cholesterol deposition and Plaque formation).
  More VLDL → more LDL



- **HDL is good** (Trap excess cholesterol and send back to liver)


Second Edition
BIOCHEMICAL
CALCULATIONS
ILLEGAL
for Sale in
USA
IRWIN H. SEGEL
WILEY
STUDENT
EDITION
RESTRICTED!
FOR SALE ONLY IN
INDIA,
WILEY



Next: Vitamins , Minerals and water